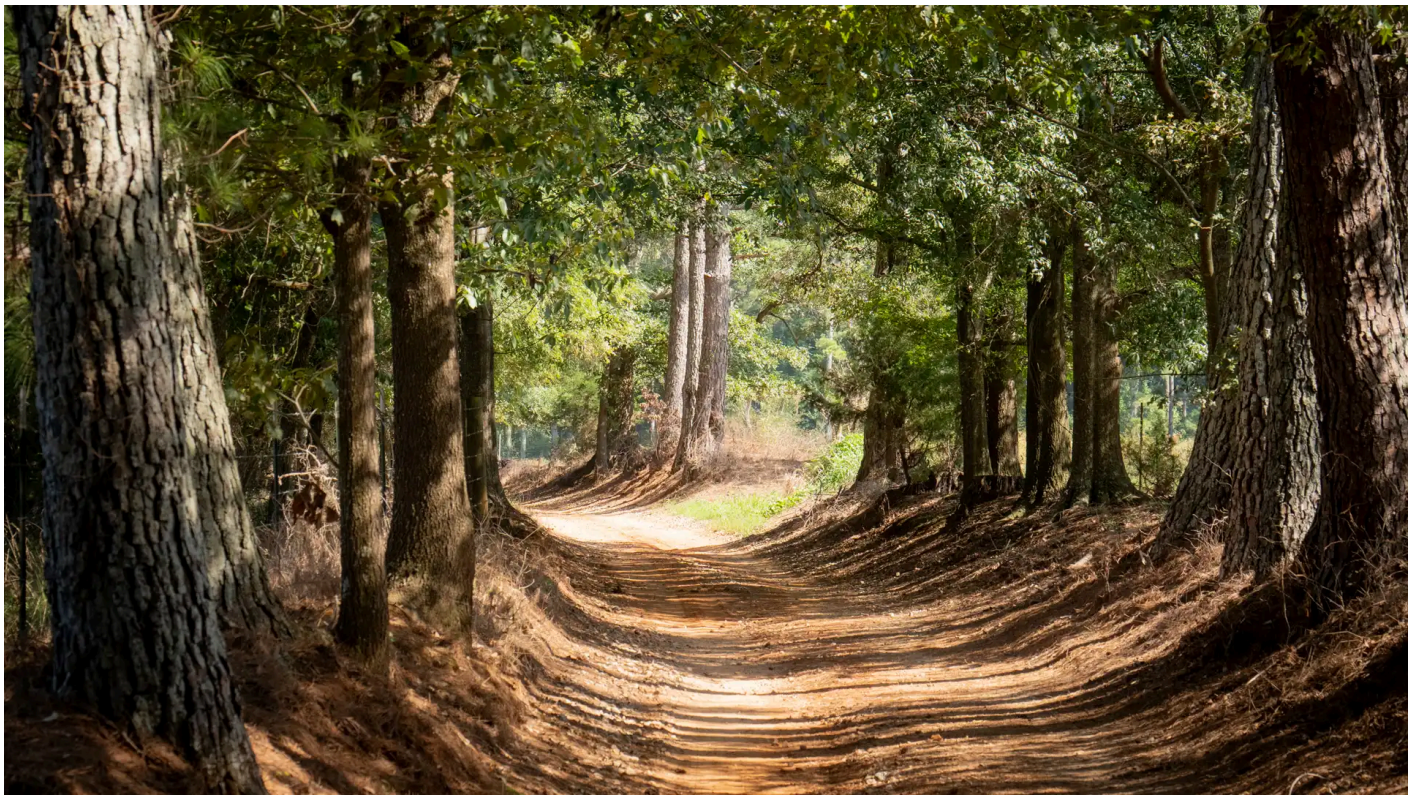
Not Your Grandpa's Farm: High Tech Boosts Output, Stewards Resources

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David Pendered
Senior Content and Publishing Specialist

- Article
- Regional Economy
- Employment



The well-worn road to Innisfail Farm. Photo by Stephen Nowland

The dirt road that leads to a high-tech cattle-feeding machine marks the time that has passed since the early days of Innisfail Farm, which was founded in the 1870s when Whitey Hunt's great-great-grandfather bought the land. Hunt can't guess when the road was cut. It had to be long ago, he said, based on the road's surface, which has been worn down a few feet beneath the road's shoulders.

Hunt's farm is part of one of Georgia's leading industries, agriculture. The state's farms, numbering just over 39,000, produced a total farm gate value of just over $18 billion in 2024, according to, respectively, the 2022 Census of Agriculture , which is the latest available census compiled by the U.S. Department of Agriculture, and the 2026 Georgia Ag Impact Report , produced by the University of Georgia. In Georgia, food and fiber production in 2024 supported 370,400 jobs and contributed more than $100 billion to Georgia's $1.5 trillion in economic input, according to UGA's report.

Keeping a watchful eye on a changing economic sector

The Federal Reserve Bank of Atlanta monitors the agriculture economy of the Southeast through the Bank's Agriculture Advisory Council. The council's 12 members offer the Atlanta Fed their views on the region's production trends across the various areas that make up the agriculture sector, such as citrus, cotton, timber, dairy, row crops, beef and poultry, and grain. "The council contributes valuable insights to the Atlanta Fed about the state of the agriculture economy in the Southeast, and this knowledge helps the Atlanta Fed inform monetary policymakers on projected economic conditions in this region's agriculture sector," said Anoop Mishra, vice president and regional executive in the Atlanta Fed's Birmingham Branch.



Members of the Atlanta Fed's Agriculture Advisory Council. Photo by Scott Taylor

The ag sector is evolving rapidly with the development of technology that supports the cultivation of plants and animals. For years, Hunt—who has specialized in the Hereford breed of cattle since the late 1970s—has used the latest technologies to guide his decisions as he grew and sold animals on Innisfail Farm, a rolling stretch of land with working forest and pastureland between Big Indian Creek and Little Indian Creek, just south of the Morgan County town of Madison, Georgia. (Hunt still tends the farm, along with his eldest son, Weyman Hunt, and Breck Debnam, the farm's herdsman, who was named 2026 Hereford Herdsman of the Year by the American Hereford Association.)



Hunt keeps a watchful eye on part of his herd of Herefords. Photo by Stephen Nowland

In 1972, when he first entered the cattle business, Hunt used a rudimentary hanging scale to weigh newborn calves, recording their weight from birth through sale. In an era when cattle were managed mostly by human eye and touch, some of his friends and competitors scoffed at Hunt's scientific approach and data collection. The practice of cattle raising hadn't changed much in the nearly two centuries, since Thomas Jefferson wrote in his Farm Book that his cattle were to be fed "2. ton of dry hay a year, or green equivlt."

The cattle business has changed a lot since the 1970s. One breakthrough in that decade was the introduction of ear tags with insecticide that ward flies off cattle, allowing animals to shift their energy from tail-swishing to weight gain. In this century, genetic testing has become widespread to better predict an animal's marbling score, tenderness, feed efficiency, and calving ease.

The evolution in cattle management is part of the broader technological advancement in the agriculture sector that enabled the United States to become the breadbasket of the world in the wake of World War II. The US Department of Agriculture's Economic Research Service (ERS) credits technology with total farm output nearly tripling from 1948 to 2023.

The ERS notes that this significant growth rate is "all the more remarkable" because of the simultaneous, and dramatic, declines in inputs for labor and land. From 1948 to 2023, labor and land inputs declined by 76 percent and 28 percent, respectively, as farmers shifted their spending toward machinery, chemicals, purchased services, and other materials, according to the ERS report.

Georgia's experience with a declining number of farm workers seems to track the national pattern. The ERS does not provide data more granular than the national level, but the USDA's Census of Agriculture provides a rough comparison that suggests farm labor in Georgia has declined precipitously. For the year 1945, the census shows that more than 314,000 individuals worked on farms; that figure includes family members and hired workers, all of them over age 14. For 2022, the census shows that almost 45,000 workers were paid to work on farms; the report does not mention the number of family members working on farms.

Combining high tech and lowing cattle

Hunt has kept pace with technology and stays current through groups such as the American Hereford Association, where he served on the board representing the Southeast.

On Innisfail Farm, the latest innovation Hunt has adopted is an automated feed system. He had his eye on the system for 20 years before buying it three years ago, a purchase he delayed in part because it was a sizable investment for his 100-head Hereford operation. For Hunt, the appeal of the system was its ability to track the amount of food each animal eats each day and the resulting daily weight gain that the system also measures. The data will serve many functions, including the identification of those animals that gain the most weight on less food so he can target them for breeding.



Innisfail Farms's Whitey Hunt. Photo by Stephen Nowland

The traditional feeding system, still in widespread use nationwide and on portions of Innisfail Farm, amounts to cattle foraging in pastures and eating hay or grain placed to serve an entire herd. In terms of resource management, this system's drawback is that it tracks only feed given to the entire herd, not the amount each animal consumes. Fast-evolving technology has addressed the cost of feed, an input that can exceed 50 percent of an operation's total expenditures. For example, one such device tracks each animal's methane production, and this information can allow farm workers to alter the composition of feed to lower the production of greenhouse gas and monitor feed efficiency.

Cattle breeding has also benefited from new technologies. For example, after a genetically desirable cow is impregnated via artificial insemination with sperm collected from a bull with desired traits, the resulting embryos can be inspected, and some are implanted into surrogate mothers for gestation, resulting in calves with the most desirable traits.

These advances in cattle management are occurring within the broader context of precision agriculture. Technology, big data, analytics, computer modeling, informatics, and the broader Internet of Things are among the next generation of tools in the quest to provide food and fiber, and their efficient transport, to the world's growing population. Just as Jefferson measured the

amount of feed provided to each animal to manage his costs to raise a herd, modern crop farmers seek to optimize their management of fields with satellite navigation systems that communicate with devices that can include:

- Drones equipped with high-resolution and thermal imaging cameras to monitor crops for signs of stress and pest infestations before they're visible to the naked eye;
- Ground robots to plant seeds, apply chemicals, and harvest crops; and
- Moisture sensors placed in the ground in fields and pastures to reduce irrigation costs and water waste by ensuring water is supplied only when and where it's needed.

Sonny Perdue has had a hand in promoting technological advances in the agriculture sector. Perdue was a former two-term Georgia governor and then became secretary of the US Department of Agriculture in the first Trump administration. And he is the current, though retiring, chancellor of the University System of Georgia. Perdue helped usher in two high-tech incubators for innovative ag technology.

As agriculture secretary, Perdue supported the establishment of Grand Farm in North Dakota. Grand Farm, a 590-acre spread operated by a subsidiary of a nonprofit entity that promotes economic growth in North Dakota, is a demonstration platform for new agricultural technologies being devised by an assortment of small business, major corporations, academicians, and entrepreneurs. In April, the Department of Agriculture announced that Grand Farm will serve as the national program manager of the newly unveiled National Proving Grounds Network for AgTech, which is to evaluate agricultural technologies in real-life farming and ranching conditions. Partners in the program include land grant institutions nationwide and other USDA research partners.

In Georgia, as chancellor, Perdue promoted the partnership of the Grand Farm and UGA College of Agricultural and Environmental Sciences to establish UGA Grand Farm , a project that promises to foster new technology as Georgia's agriculture industry faces a mounting number of challenges. One of them is the call from state policymakers to reduce the amount of water used for crop irrigation, as identified in the state's long-range Comprehensive Statewide Management Plan and the updated 2025 implementation plan. To this end, a product tested at the UGA Grand Farm in 2025 aims to help soil retain more water for longer periods, reducing the amount of water needed for irrigation.

Another goal of the program is to encourage small operators, ones like Whitey Hunt, to employ the latest technology in the hope that their yields will benefit. A majority of Georgia's farms are significantly smaller than the national average, with 73 percent of Georgia farms covering 179

acres or less, according to the UGA report. In 2025, the national average for almost 1.9 million farms was 469 acres, according to a report in February by the USDA's National Agricultural Statistical Service.

The collaboration between the Grand Farm and UGA was announced in 2024 and launched in 2025 on a tract of 250 acres, located near the Georgia National Fairgrounds, in the central Georgia city of Perry. The program's first year saw five inaugural field projects tested, including:

- An underground system to monitor nitrogen, nitrate, moisture, and more;
- Seed coatings;
- A microparticle delivery technology that aims to improve delivery of biologicals to seeds, leaves, and fertilizer;
- A polymer made of fruit residue, not petroleum, to gradually release water into the soil (polymers already are used to slow-release fertilizer); and
- A living soil amendment that promotes drought resistance, absorbs carbon dioxide, and improves nitrogen, phosphorus, and micro-nutrients.

Planting the seeds for tomorrow

In Madison, Whitey Hunt has a reputation for helping others. That extends to offering a hand to novice farmer Emily Rom, who's taking over the family-owned Hayes Heritage Farm a few miles north of Hunt's farm. Hunt said he wants to help Rom partly because she reminds him of himself at her age, and in part to pay it forward. "I had a great mentor who worked for our family, a great person. He ran the [family-owned] feed mill for 45, 48 years," Hunt said. "I had a lot of respect for him, and for everything he knew and told me, and his work ethic."

Rom said she could benefit from a mentor who could help her navigate the complexities of cattle farming. She has a bachelor's degree in business management but not the type of education that would help her manage Hayes Heritage Farm. "I'm fifth generation on this farm, and I have respect for the generations of my family who worked here before me," she said. "My grandfather inspired that, and I don't want his work to go to waste." But the task is daunting. Almost everywhere she turns, Rom faces a potential pitfall. For example, some species of grass in her pasture can be toxic to cattle during a drought because drought promotes production of hydrogen cyanide in the grass. Another challenge involves a bull that contracted pinkeye, a highly contagious disease associated with dry grass and face flies and which reduces the animal's value for breeding. The farm isn't turning a profit despite Rom's 70-hour workweeks and no vacation in years.



Hayes Heritage Farm's Emily Rom. Photo by Stephen Nowland

Rom isn't alone in feeling the strain of farming for a living. Georgia farmers are selling their property at a rapid pace. Between 2001 and 2016, 544,000 acres of agricultural land in Georgia was converted from agricultural use to residential or commercial development, the fourth-highest amount of agricultural land sold in any state, according to a report by the American Farmland Trust, a nonprofit entity that seeks to conserve land for agricultural use. Georgia in 2024 had 9.9 million acres of land used by farms, almost a third of it in harvested cropland, according to UGA's Ag Impact Report.

Perdue gave Rom words of encouragement when he visited during a tour of area farms hosted by the Madison-Morgan Conservancy, an environmental conservation organization. Rom appreciated Perdue's vote of confidence and said she needs someone she can consult on a regular basis.



Sonny Perdue and Emily Rom. Photo by Stephen Nowland

Hunt said he's been in Rom's shoes. He almost quit Innisfail Farm early in his career on the farm after losing 19 calves out of 21 calves delivered. The culprit was a bacterium, leptospirosis, which is spread in the urine of infected animals and, sure enough, Hunt had been keeping the pregnant cows in an old chicken barn he thought would provide shelter but proved to be a breeding ground for the fatal bacteria. Like Rom with her challenges, Hunt didn't have the experience or guidance to know that pregnant cows should not be kept in a place where urine and manure mix with the soil where cows stand. Hunt thinks he can help Rom avoid similar pitfalls, noting that he has a few words to help when spirits sag: " 'Be more efficient with time, if you can. Get help when you need it, if you can.' I think she can do it."